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Lexical Semantic Change Detection in Computational Linguistics: An Overview

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Word Usage Graphs

Unrecorded Sense Detection

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Introduction

- ▶ Lexical Semantic Change Detection
 - ▶ **goal:** automate the analysis of changes in word meanings over time
 - (1) ***Mäuse** und Ratten sind selbstverständlich mit den europäischen Schiffen auch hierher gekommen.*
'Of course, **mice** and rats also came here with the European ships.'
 - (2) *Deshalb eignet sich die **Maus** auch für verschiedene Betriebssysteme neben Windows und macOS.*
'The **mouse** is therefore also suitable for various operating systems in addition to Windows and macOS.'
- ▶ applications in historical linguistics, lexicography, but also social sciences . . .

Lexical Semantic Change

- ▶ in historical semantics word meaning change is generally defined as changes in word senses
- ▶ a common definition is Blank (1997)'s distinguishing two main types:
 - ▶ **innovative meaning change**: emergence of a full-fledged additional sense of a word
 - ▶ **reductive meaning change**: loss of a full-fledged sense of a word

The process

- ▶ innovative meaning change begins in language discourse
- ▶ **semantic innovation**
- ▶ neglected or adopted by other speakers
- ▶ **lexicalization**

Semantic Innovations

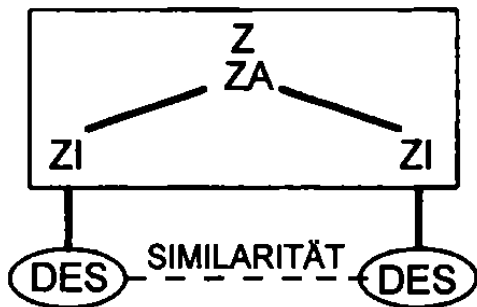
- ▶ occur because of psychological associations
- ▶ name new referents with names of associated referents

The case of *Maus*

engl. *mouse*

'kleines Nagetier'

'graph. Zeigegerät'



(länglich-runde Form,
langer Schwanz)

(länglich-runde Form,
schmales, langes Kabel)

The case of *Maus*

- (1) ***Mäuse** und Ratten sind selbstverständlich mit den europäischen Schiffen auch hierher gekommen.*

‘Of course, **mice** and rats also came here with the European ships.’

- (2) *Deshalb eignet sich die **Maus** auch für verschiedene Betriebssysteme neben Windows und macOS.*

‘The **mouse** is therefore also suitable for various operating systems in addition to Windows and macOS.’

The case of *Presse*

- (1) Der zweyte Theil vom Bauernrechte ist schon lange aus der **Presse**;
*'The second part of Farmers' Rights already left the **press**;*
- (2) *Alle Freiheiten suspendirt! die persönliche Freiheit wie die der **Presse**!*
*'All freedoms suspended! the personal freedom as well as the one of the **press**!'*

The case of *Zufall*

- (1) [...] daß sie aus Angst **Zufälle** bekommen und wieder gefährlich krank werden würde.
*'[...] that she would have **seizures** out of fear and become dangerously ill again.'*
- (2) *Es muß verbrannt sein, vielleicht durch **Zufall**, vielleicht durch Feindeshand.*
*'It must have burned, perhaps by **coincidence**, perhaps by enemy hands.'*

Two measurement paradigms

1. Word Usage Graphs

(Schlechtweg, 2023)

- ▶ compares **corpus** to **corpus**
- ▶ builds on **Word Sense Induction**

(Schütze, 1998)

2. Unrecorded Sense Detection

(Erk, 2006; Fedorova et al., 2024)

- ▶ compares **corpus** to **dictionary**
- ▶ builds on **Word Sense Disambiguation**

(Navigli, 2009)

Word Usage Graphs

Human Measurement of Lexical Semantic Change

A	1824	and taking a knife from her pocket, she opened a vein in her little arm ,	😊
B	1842	And those who remained at home had been heavily taxed to pay for the arms , ammunition;	✖
C	1860	and though he saw her within reach of his arm , yet the light of her eyes seemed as far off	😊
		...	
D	1953	overlooking an arm of the sea which, at low tide, was a black and stinking mud-flat	🗿
E	1975	twelve miles of coastline lies in the southwest on the Gulf of Aqaba, an arm of the Red Sea.	🗿
F	1985	when the disembodied arm of the Statue of Liberty jets spectacularly out of the	😊

Table 1: Sample of diachronic corpus.

Word Use Pairs

- (A) [...] and taking a knife from her pocket, she opened a vein in her little **arm**, and dipping a feather in the blood, wrote something on a piece of white cloth, which was spread before her. 😊
- (D) It stood behind a high brick wall, its back windows overlooking an **arm** of the sea which, at low tide, was a black and stinking mud-flat [...]

Semantic Proximity Scale



4: Identical
3: Closely Related
2: Distantly Related
1: Unrelated

Table 2: DUREl relatedness scale.

Graph representation

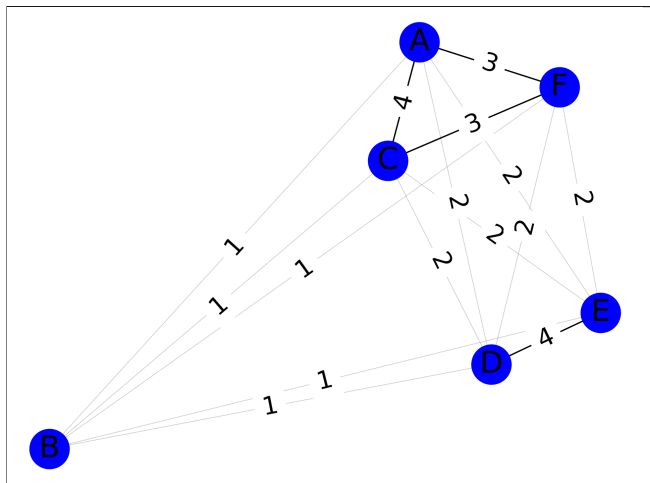


Figure 1: Word Usage Graph of English *arm*.

Clustering

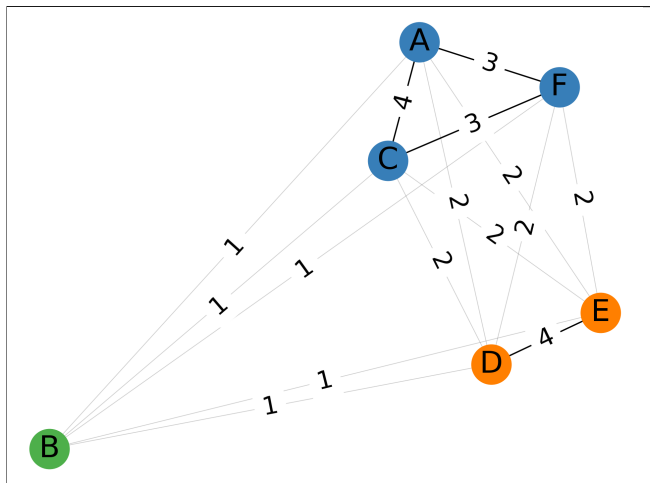


Figure 2: Word Usage Graph of English *arm*. $D = (3, 2, 1)$.

Lexical Semantic Change

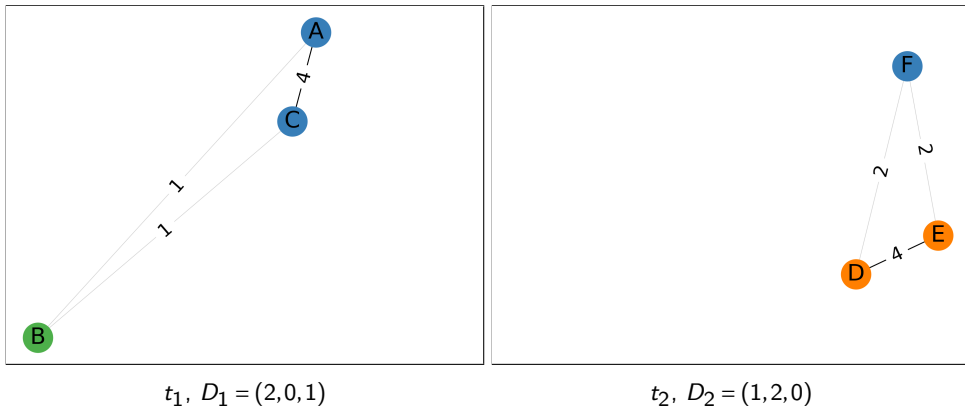


Figure 3: $B(w) = 1$ and $G(w) = 0.74$

COMPARE subgraph

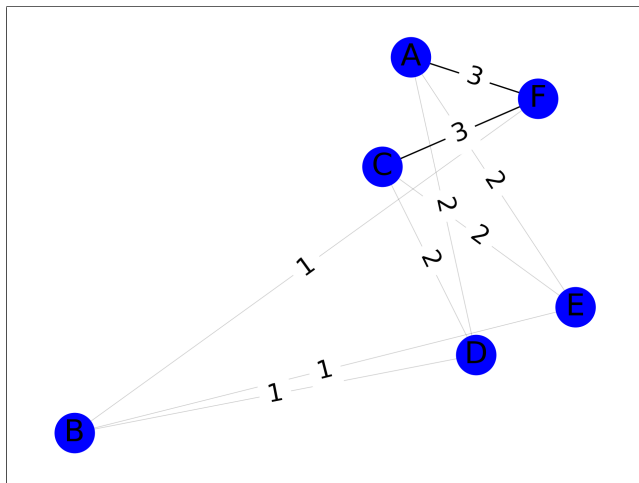


Figure 4: COMPARE subgraph of English *arm*.

$$W_{1,2} = \{3, 3, 2, 2, 2, 2, 1, 1, 1\}. \quad C(w) = 1.89$$

Change Scores

- ▶ **binary change** (loss and gain of senses)
- ▶ **graded change** (changes in sense probabilities)
 - ▶ **COMPARE score** (average of annotations between time periods)

Formal Definitions

- ▶ $B(\mathbf{D}, \mathbf{E}) = 1$ if for some i , $D_i \leq k$ and $E_i \geq n$, or vice versa.
 $B(\mathbf{D}, \mathbf{E}) = 0$ else.

- ▶ $G(\mathbf{P}, \mathbf{Q}) = JSD(P, Q) = \sqrt{\frac{KLD(P \parallel M) + KLD(Q \parallel M)}{2}}$ where
 $KLD(P \parallel Q) = \sum_i^K p_i \log_2 \left(\frac{p_i}{q_i} \right)$
 $M = \frac{P+Q}{2}$

- ▶ $C(\mathbf{W}_{1,2}) = -\frac{1}{|W_{1,2}|} \sum_{w \in W_{1,2}} w$ where
 $W_{1,2}$ are all the weights in the COMPARE subgraph.

Evaluation Tasks

- Task 1 Binary classification:** for a set of target words, predict the binary change score (evaluated by F1, precision, recall, accuracy)
- Task 2 Ranking:** rank a set of target words according to their graded change score (evaluated by Spearman correlation coefficient)

(Schlechtweg et al., 2020)

Example: Swedish *ledning*¹

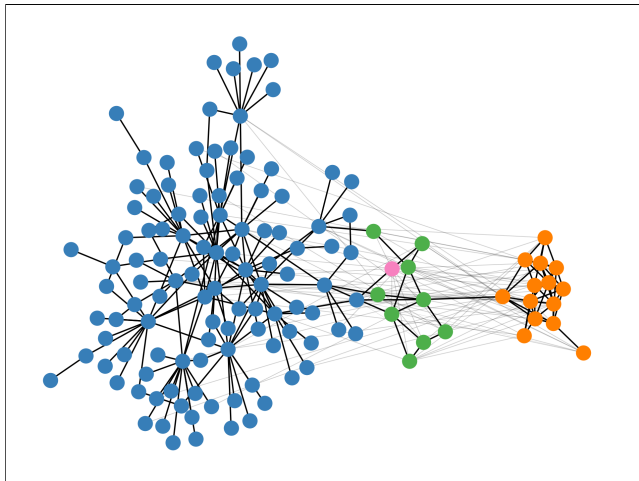


Figure 5: WUG of Swedish *ledning*.

¹Datasets available at <https://www.ims.uni-stuttgart.de/data/wugs>

Example: Swedish *ledning*

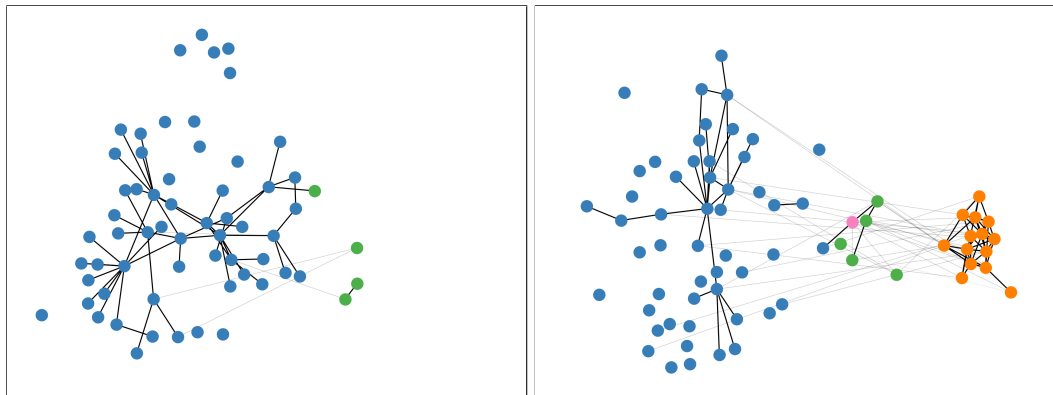


Figure 6: WUGs of Swedish *ledning*: subgraphs for 1st time period G_1 (left) and 2nd time period G_2 (right). $D_1 = (58, 0, 4, 0)$, $D_2 = (52, 14, 5, 1)$, $B(w) = 1$ and $G(w) = 0.34$.

Example: German *Eintagsfliege*

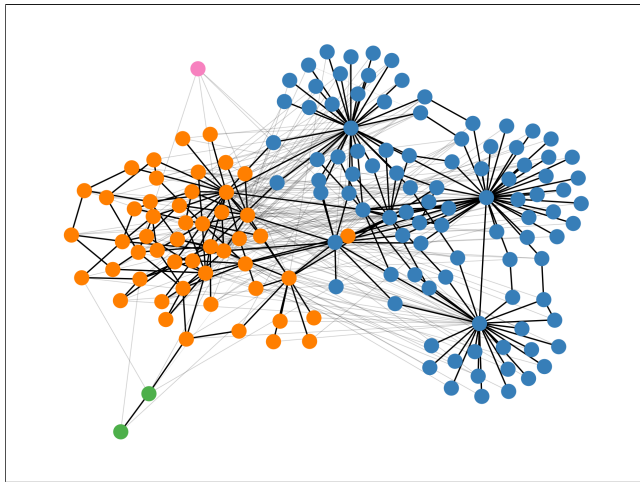


Figure 7: WUG of German *Eintagsfliege*.

Example: German *Eintagsfliege*

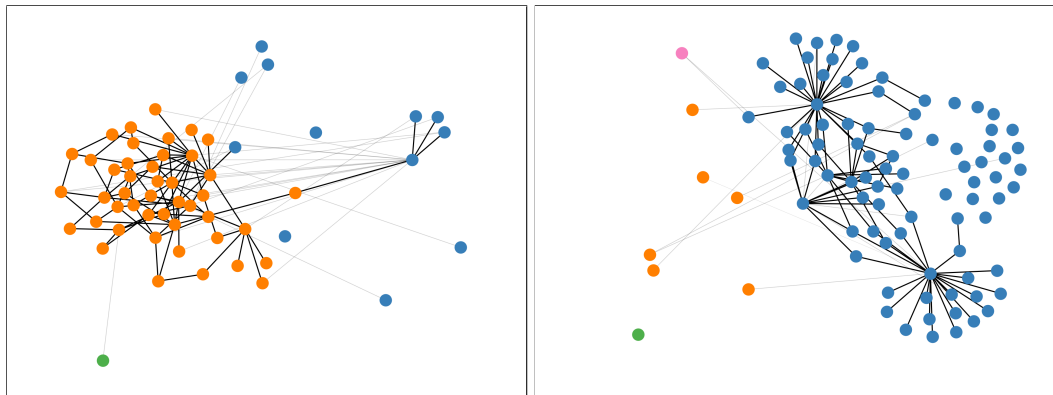


Figure 8: WUG of German *Eintagsfliege*: subgraphs for 1st time period G_1 (left) and 2nd time period G_2 (right). $D_1 = (12, 45, 0, 1)$, $D_2 = (85, 6, 1, 1)$, $B(w) = 0$ and $G(w) = 0.66$.

Summary of Annotation Steps

1. semantic proximity labeling
2. clustering
3. change measurement

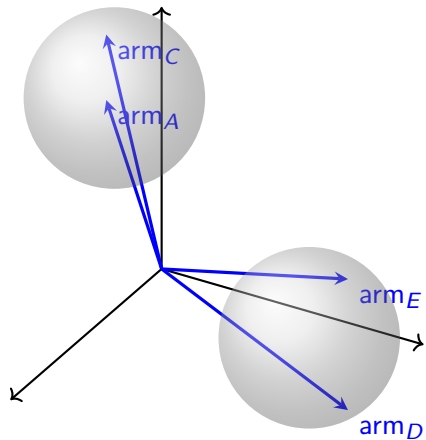
Summary of Annotation Steps with Tasks

1. semantic proximity labeling ↔ **Word-in-Context Task**
2. clustering ↔ **Word Sense Induction**
3. change measurement ↔ **Lexical Semantic Change Detection** (including previous tasks)

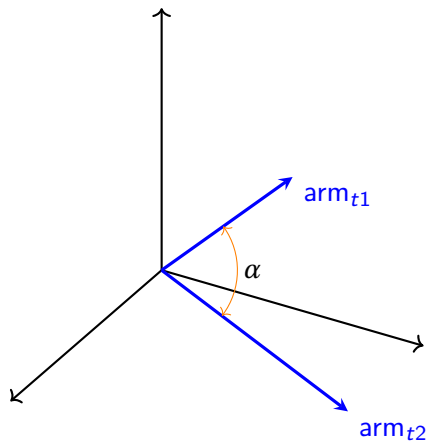
Computational Measurement of Lexical Semantic Change

- ▶ Typical **token**-based model is composed by
 1. semantic proximity model (e.g. similarity between contextualized embeddings)
 2. clustering method (optional)
 3. change measure
 - ▶ model the human measurement process
 - ▶ one vector per word use (BERT, ELMo)
- ▶ Typical **type**-based model is composed by
 1. semantic representation per word (type vector)
 2. alignment
 3. measure
 - ▶ do **not** model the human measurement process
 - ▶ one average vector per word (Word2Vec, GloVe)

Simple token-based Model



Simple type-based Model



SOTA Model Components

- ▶ use fine-tuned semantic proximity base model
- ▶ some clustering algorithm, or none
- ▶ a suitable change measure

Semantic Proximity Models

- ▶ aka **Word-in-Context** models (Pilehvar & Camacho-Collados, 2019)
- ▶ estimate degree of semantic proximity/same-sense probability for two input texts
- ▶ training data example:
 - (A) [...] and taking a knife from her pocket, she opened a vein in her little **arm**, and dipping a feather in the blood, wrote something on a piece of white cloth, which was spread before her. 😊
 - (D) It stood behind a high brick wall, its back windows overlooking an **arm** of the sea which, at low tide, was a black and stinking mud-flat [...]. 🚫
label: 0 (binary), 2 (ordinal)
- ▶ SOTA: DeepMistake, XL-Lexeme, XL-DUReL, GlossReader (Arefyev et al., 2021; Arefyev & Rachinskiy, 2021; Cassotti et al., 2023; Yadav & Schlechtweg, 2025)
- ▶ optimized on binary or ordinal multilingual semantic proximity data (Martelli et al., 2021; Pilehvar & Camacho-Collados, 2019; Schlechtweg et al., 2025)
- ▶ have brought **major performance improvements** (Kutuzov & Pivovarova, 2021)

Semantic Proximity Models

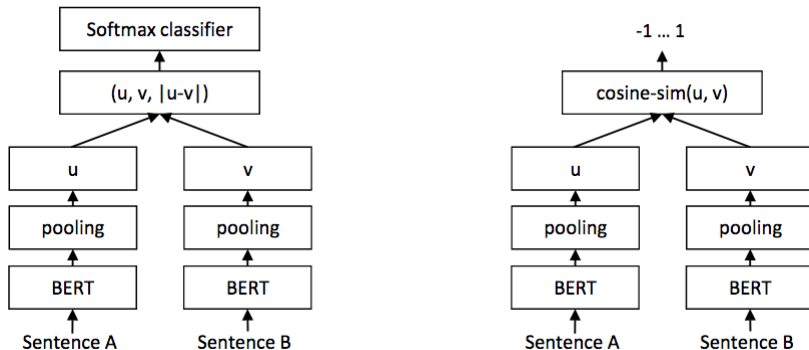


Figure 9: S-BERT (Reimers & Gurevych, 2019) training architecture used for XL-Lexeme and XL-DURel.

Word Sense Induction Models

- ▶ aka **clustering**
- ▶ wealth of algorithms available
- ▶ Agglomerative, Spectral, Correlation, Stochastic Blockmodel
(cf. Schlechtweg, Zamora-Reina, et al., 2024)
- ▶ **default:** Correlation Clustering (Bansal et al., 2004)
 - ▶ straightforward because used for ground truth clustering
 - ▶ **additional advantages:** finds number of clusters, intuitive, interpretable

Correlation Clustering

- ▶ W is given by semantic proximity predictions from graph $G = (U, E, W)$
- ▶ $w \in W$ are shifted to obtain a set of **positive** and **negative** edges
- ▶ Let $C : U \mapsto L$ be some clustering on U
- ▶ $\phi_{E,C}$ is the set of positive (high) edges **across** any of the clusters in clustering C
- ▶ $\psi_{E,C}$ the set of negative (low) edges **within** any of the clusters
- ▶ correlation clustering searches for a clustering C that minimizes the sum of weighted cluster disagreements:

$$SWD(C) = \sum_{e \in \phi_{E,C}} W(e) + \sum_{e \in \psi_{E,C}} |W(e)| .$$

- ▶ **main assumption:**
 - ▶ weights above the threshold indicate same sense, below the threshold they indicate different sense

Evaluation

- ▶ Adjusted Rand Index (ARI) (Hubert & Arabie, 1985)
 - ▶ accuracy on pairwise cluster agreements between nodes
 - ▶ controlled against agreement by chance
 - ▶ 1.0 means perfect accuracy, 0.0 means accuracy expected by chance

Change Measures

- ▶ **binary change:**

- ▶ cluster gain/loss (cluster-based)
- ▶ thresholding graded predictions

(Schlechtweg et al., 2020)

- ▶ **graded change:**

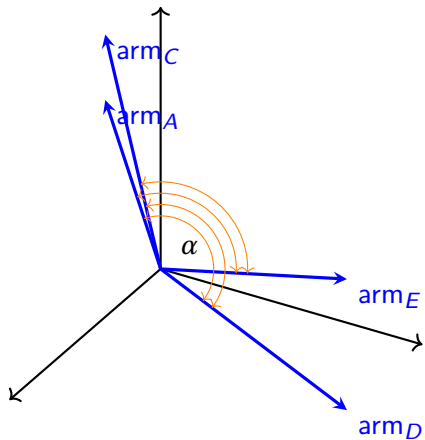
- ▶ Jensen Shannon Distance (cluster-based)
- ▶ Average Pairwise Distance

(Lin, 1991)

(Kutuzov & Giulianelli, 2020)

- ▶ cluster-based vs. summary-based
- ▶ exact models vs. not
- ▶ summary-based dominates for graded change

SOTA Model for graded change: APD



Results

Lang.	Binary	Model	Graded	Model
Chinese			.73	XL-Lex.+APD
English	.70 (.67/.75)	BERT+HDBSCAN	.89	XL-Lex.+APD
German	.70 (.60/.82)	SGNS+thres.	.84	XL-Lex.+APD
Norwegian			.76	XL-Lex.+PRT
Russian			.86	XL-Lex.+APD
Swedish	.64 (.47/1.0)	XLM-R+K-means	.81	XL-Lex.+APD
Spanish	.72 (.62/.86)	GlossR.+APD+thres.	.74	GlossR.+APD

Table 3: SOTA performances on LSCD tasks (Cassotti et al., 2023; Periti & Tahmasebi, 2024a; Rachinskiy & Arefyev, 2022; Schlechtweg et al., 2020; Zamora-Reina et al., 2022). Values give F1 (P/R) for binary change and Spearman for graded change. XL-DUREl left out as not comparable.

Summary

- ▶ **advantages:**

- ▶ no need for sense definitions
- ▶ rather explicit annotation criteria

- ▶ **disadvantages:**

- ▶ questionable notion of semantic proximity
- ▶ quadratically increased annotation load
- ▶ need for clustering algorithm

(Schlechtweg, Cassotti, et al., 2024)

- ▶ **open questions:**

- ▶ clustering on gold data
- ▶ cluster models & binary change
- ▶ application
- ▶ multiple time periods
- ▶ types of change
- ▶ detect noisy usages
- ▶ error analysis

(Graef, 2025)

(Sköldbberg et al., 2024)

(Periti & Tahmasebi, 2024b)

(Whaley, 2025)

(Choppa et al., 2025)

DURel tool²

- ▶ demo
- ▶ register, upload, data inspection, annotation, visualization, computational annotation

²<https://durel.ims.uni-stuttgart.de>

WUG datasets³

- ▶ demo
- ▶ website, code repository, folder structure
- ▶ some examples

³<https://www.ims.uni-stuttgart.de/data/wugs>

graft

- ▶ **lemma:** graft
- ▶ **POS:** noun
- ▶ **dataset:** dwug_en_resampled
- ▶ **senses⁴:**
 - ▶ "shoot inserted into another plant," late 15c
 - ▶ "corruption," 1865, perhaps 1859

⁴[etymonline.com](https://www.etymonline.com)

- ▶ **lemma:** stab
- ▶ **POS:** noun
- ▶ **dataset:** dwug_en_resampled
- ▶ **senses⁵:**
 - ▶ mid-15c., stabbe, "wound produced by stabbing," from or related to stab (v.).
 - ▶ The meaning "act of stabbing, thrust or blow with the point of a weapon" is from 1520s.
 - ▶ The meaning "a try" is recorded by 1895, American English, probably on the notion of "a vigorous thrust."
 - ▶ The verb was long used in reference to a wound given treacherously; stab in the back in the figurative sense "treacherous deed" is attested by 1881; the verbal phrase in the figurative sense is from 1888.

⁵etymonline.com

Exercise

- ▶ **Task:** Find out whether and how word **record** has changed its meaning over time. Build two groups: one group inspects the raw data, one the human annotation.

record

- ▶ **lemma:** record
- ▶ **POS:** noun
- ▶ **dataset:** dwug_en_resampled
- ▶ **senses⁶:**
 - ▶ c. 1300, "testimony committed to writing, fact or condition of having been recorded,"
 - ▶ "a written account of some fact, event, or proceeding for the purpose of preserving the memory of it" is from late 14c.
 - ▶ "disk on which sounds or images have been recorded" is attested from 1878
 - ▶ "best or highest official achievement in a sport, activity, etc." is by 1883
 - ▶ The journalist's phrase on the record is attested from 1900; adverbial phrase off the record "confidentially" is attested from 1906.

⁶etymonline.com

The LSCD Benchmark

- ▶ Lexical Semantic Change Detection (Schlechtweg, 2023)
- ▶ **heterogeneity** and **modularity** in models, datasets and tasks
- create one repository⁷ standardizing model component combinations, dataset preprocessing and evaluation

⁷<https://github.com/Garrafao/LSCDBenchmark>

Usage Example

```
python main.py \  
  dataset=dwug_de_210 \  
  dataset/split=dev \  
  dataset/preprocessing=raw \  
  task/lscd_graded@task.model=apd_compare_all \  
  task/wic@task.model.wic=contextual_embedder \  
  task/wic/metric@task.model.wic.similarity_metric=cosine \  
  task.model.wic.ckpt=bert-base-german-cased \  
  task=lscd_graded \  
  evaluation=change_graded
```

Compare graded change models

- ▶ **Average Pairwise Distance (APD)**: estimates the edge weights from COMPARE graph and takes their mean: $G(w) = \text{mean}(\text{COMPARE})$ (Kutuzov & Giulianelli, 2020)
- ▶ **DiaSense**: normalizes APD by weights from full graph:
 $G(w) = \text{mean}(\text{COMPARE}) - \text{mean}(\text{FULL})$ (Beck, 2020)

Benchmark command APD

```
python main.py \  
  dataset=dwug_de_210 \  
  dataset/split=dev \  
  dataset/preprocessing=toklem \  
  task/lscd_graded@task.model=apd_compare_sampled \  
  task/wic@task.model.wic=contextual_embedder \  
  task/wic/metric@task.model.wic.similarity_metric=cosine \  
  task.model.wic.ckpt=bert-base-german-cased \  
  task=lscd_graded \  
  evaluation=change_graded
```

Benchmark command DiaSense

```
python main.py \  
  dataset=dwug_de_210 \  
  dataset/split=dev \  
  dataset/preprocessing=toklem \  
  task/lscd_graded@task.model=diasense_sampled \  
  task/wic@task.model.wic=contextual_embedder \  
  task/wic/metric@task.model.wic.similarity_metric=cosine \  
  task.model.wic.ckpt=bert-base-german-cased \  
  task=lscd_graded \  
  evaluation=change_graded
```

Some Research Questions

- ▶ How do models perform on new data sets?
- ▶ Are current results for SOTA models reproducible on cleaned data?
- ▶ Can we find better measures for graded change than APD?
- ▶ Can clustering on optimized WiC models improve results on binary change?
- ▶ How do current models for binary change perform in scenarios where correlation is low with graded change?
- ▶ How do current models for graded change perform in high-polysemy scenarios?
- ▶ How do model hyper-parameters generalize between data sets?

Unrecorded Sense Detection

Human Measurement of Unrecorded Senses

- A and taking a knife from her pocket, she opened a vein in her little **arm**,
- B And those who remained at home had been heavily taxed to pay for the **arms**, ammunition;
- C and though he saw her within reach of his **arm**, yet the light of her eyes seemed as far off
- D overlooking an **arm** of the sea which, at low tide, was a black and stinking mud-flat
- E twelve miles of coastline lies in the southwest on the Gulf of Aqaba, an **arm** of the Red Sea.
- F when the disembodied **arm** of the Statue of Liberty jets spectacularly out of the

headword	sense definition
arm	a human limb
	weapon system

Figure 10: Sample corpus (left) and dictionary entry (right).

Use-Sense Comparison

(D) It stood behind a high brick wall, its back windows overlooking an **arm** of the sea which, at low tide, was a black and stinking mud-flat [...]

Sense 1 a human limb

Sense 2 weapon system

Sense 3 [NEW SENSE]

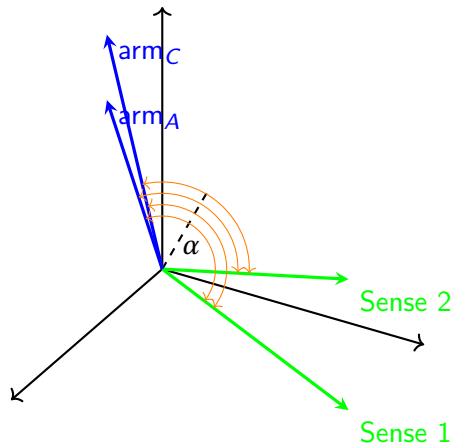
Evaluation tasks

- ▶ recent **shared task** introduces two evaluation tasks (Fedorova et al., 2024)
- ▶ **data**: Finnish, Russian, German
- ▶ **task**: given a set of historical reference use-sense pairs, for a set of usages...
 1. **unrecorded sense detection**
 - ▶ disambiguate usages belonging to covered senses (evaluated by F1)
 - ▶ cluster a set of usages into senses (evaluated by ARI)
 2. **sense definition generation**
 - ▶ generate a sense definition for uncovered senses (evaluated by BERTScore/BLEU)

Computational Measurement of Unrecorded Senses

- ▶ Typical model is composed by
 1. semantic proximity model (this time between uses and senses)
 2. binary unrecordedness classifier
 3. clustering method (optional)
 4. sense generation model (optional)

Simple Model: Threshold on Cosine Similarity



SOTA Model Components

► SOTA Model used for the different levels:

1. **semantic proximity**: GlossReader
2. **classifier**: logistic regression on vector distance features
3. **clustering**: Agglomerative

(Kokosinskii et al., 2024)

(Arefyev & Rachinskiy, 2021)

Results

Lang.	F1	Model	ARI	Model
German	.76	GlossR.	.54	GlossR.
Finnish	.76	GlossR.	.65	GlossR.+LR+Agglom.
Russian	.75	GlossR.	.26	GlossR.+Agglom.

Table 4: SOTA performance on Unrecorded Sense Detection task (Fedorova et al., 2024; Kokosinskii et al., 2024).

Computational Generation of Sense Definitions

- ▶ typically a **large language model** (with decoder) (Open AI, 2022; Touvron et al., 2023)
- ▶ i.e., GPT, Llama, DeepSeek
- ▶ **pre-trained** on large amounts of text data with language modeling task
- ▶ optionally **fine-tuned** with specific task instructions
- ▶ often requires **prompt engineering**

A prompt

[Instruction]:

Imagine that you are a lexicographer, given a headword {target_word} in {lang}, write the dictionary definition of its usage in the following quotations:

1. First quotation
2. Second quotation

[Requirements]:

The definition you create should be brief. A maximum of ten words is allowed. The definition ends at the first period.

[Response]:

Definition (string): {definition}

Figure 11: An illustration of Ma et al. (2024)'s prompt used to instruct GPT-3.5-turbo to generate dictionary-like definitions, where 'quotation' is synonymous of 'word usage'.

Results

Lang.	BLEU + BERTScore	Model
German	.33	unk.
Finnish	.41	GPT-3.5
Russian	.73	WSD

Table 5: SOTA performance on Generation of Sense Definitions (Fedorova et al., 2024; Kokosinskii et al., 2024).

Examples

- ▶ **usage:** *Якобы тут, возле Черниховки, находится радиоактивный могильник.*
'Allegedly here, near Chernikhovka, there is radioactive **burial ground**.'
- ▶ **definition:** *спец. место захоронения радиоактивных отходов; специальное сооружение для такого захоронения.*
'Special burial site for radioactive waste; special structure for such burial.'
- ▶ **generated definition by GPT-3.5-turbo:** *Могильник - место захоронения радиоактивных отходов или погибших.*
'Burial ground - a burial place for radioactive waste or dead.'
- ▶ **Evaluation:** BLEU: 21.2, BERTScore: 0.79

Examples

prompt: . . . The representative of the United States Government speaks about seeds for rice with, increased yield which the United States will send to Vietnam," Mr. Thuy said, "but even at this moment American troops and American **planes** are attacking our harvest, are now destroying the economic infrastructure of South Vietnam." What is the definition of **plane**?

Definition by mT0: An aircraft.

Definition by GPT-3.5-turbo: An aircraft that is used for military purposes, typically for combat or warfare.

Definition by GPT-o1-preview: An airplane; a powered flying vehicle with fixed wings and a weight greater than that of the air it displaces, used for transportation or military purposes.

Definition in OED: An aeroplane.

Analysis

- ▶ major problem: too general definitions

prompt: It follows, that if the point in space be upon the horizontal **plane**, its distance with regard to this last named **plane** will be zero or nothing, and the vertical A a will be zero also. What is the definition of **plane**?

Definition by mT0: A surface without outline; a flat or almost flat surface.

Definition in OED: A flat geometrical surface which has the property that every straight line joining any two points of the surface lies wholly in the surface.

Summary

- ▶ **advantages:**
 - ▶ dictionaries are rich resources
 - ▶ practical relevance
 - ▶ change predictions become more explainable
- ▶ **disadvantages:**
 - ▶ need for sense definitions
 - ▶ inconsistent criteria to create these
- ▶ **open questions:**
 - ▶ more straightforward evaluation
 - ▶ interpretable evaluation for generation
 - ▶ error analysis
 - ▶ application
 - ▶ exploit sense-annotated usages

(Kutuzov et al., 2024)

Applications

Applications

▶ Word Usage Graphs

- ▶ detect strongly changing words in German historical corpora with efficient type-based approaches (Kurtyigit et al., 2021)
- ▶ detect unrecorded senses in Swedish dictionary by comparing induced corpus sense number to dictionary sense number (Sander et al., 2024; Sköldberg et al., 2024)

▶ Unrecorded Sense Detection

- ▶ detect unrecorded senses in English and Swedish dictionary using Word-in-Context models (Lautenschlager et al., 2024)
- ▶ fine-tune LLMs for definition generation in English, German, Russian, Norwegian (Kutuzov et al., 2024)
- ▶ apply SOTA methods to prioritize dictionary update proposals in ODE (Blessing et al., 2026)

Conclusion

- ▶ two measurement paradigms in LSCD
 - ▶ **Word Usage Graphs** (WSI-inspired)
 - ▶ high performance on **graded change** detection
 - ▶ low performance on **binary change**, no solid evaluation
 - ▶ **Unrecorded Sense Detection** (WSD-inspired)
 - ▶ unclear performance, especially in production setting
- ▶ many **open questions**:
 - ▶ How to infer senses reliably on the gold data?
 - ▶ How to model sense clusters more closely?
 - ▶ How to evaluate unrecorded sense detection methods?
 - ▶ How to apply models efficiently and effectively in production settings?

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